

Scala (Data Science Tools & Techniques)

Unit – I: Introduction to Scala Language

Introduction: Categories of Scala, Object Oriented and Functional Programming, Why Learn Scala, Scala & Java, Apache Spark & Scala, Getting to Know Apache Spark, Scala's Performance Benefits
Installing Scala: Checking Scala Installation Status in Your System, Checking Scala Installation Status in Your System, Installing the Oracle JDK, Installing Scala on Windows, Testing and Running the Scala Commands
Using the Scala Shell: Hello World in Scala REPL, Split function, Paste mode in Scala REPL, The Auto Completion Feature, Exiting from Scala REPL

Unit – II: Variables

Introduction, Immutability of Variables, Define mutable and immutable variables, Why immutability in Scala, Mutability and type safety, Specifying types for variables, Scala Identifier Rules and Naming Conventions
Data Types: Introduction, Scala Type Hierarchy, The Boolean Type, The String Type, Multiline Strings, String Operations, String Interpolation, Length of a String, Splitting Strings, Extracting Parts of the string, Index of Characters in a String
Special Types in Scala: The Unit Type, The Any Type, Type Casting in Scala
Control Statement: Introduction, Boolean Expressions, Using {} After if/else, Nested If/Else Statements, If/Else as a Ternary Operator, Pattern Matching

Unit – III: Code Blocks, Functions & Collections in Scala

Code Blocks: Introduction and Structure of Code Block
Function: Why use functions in scala, Understanding Functions in Scala, Define a Function, Invoking a function, Type Inference, Return Statements, Function with multiple parameters, Positional Parameters, Default value of parameters in functions, Function with No Argument, Single Line Function, Passing Functions as Argument, Anonymous Functions
Collections: Introduction, Understanding Lists, Indexing List Elements, List Size, Basic Statistics, Convert a list to string, Iterating Over Lists, map Function to Iterate Over Lists, foreach for list, filter function for list, Reduce operations on list, List Equality Check, Alternate ways to Create List, Create Set, Map Collections, Indexing a Map, Uniqueness of keys in Map, Alternate ways to create Map collection, Manipulating Maps, Understanding Tuples, Indexing Tuples, Iterating over Tuples, Alternate ways to create a Tuple, Mutable Collections, Mutable Maps, Nested Collection

Unit – IV: Loop, Classes & Packages, Exception Handling

Loops: Introduction, Types of Loops in Scala, The for loop, The while Loop, Breaking a loop iteration
Classes & Packages: Introduction, Classes and Objects in Scala, Creating Classes and Objects in Scala, Singleton Objects, Use Autocompletion Tab, Case classes, Equality Check, Classes and Packages, Avoid Name Space Collusion, Importing Packages
Exception Handling: What is an exception, Exception handling in Scala, Try Catch Block, The Throwing Exceptions, Try, Catch and Finally